

MEIYARASU K

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with **2 years** of experience building production computer vision and LLM-powered systems end-to-end. Shipped **2 production CV systems** (ANPR, Face Recognition with anti-spoofing) at **~20 FPS** real-time inference and curated **100K+ image** pipelines from scratch, alongside self-driven projects in violence detection, dataset-annotation tooling, and LLM fine-tuning. Built an **Anthropic Claude-powered natural-language-to-SQL reporting agent** across a production database of **220+ tables**, with retrieval-based table selection, safety guardrails, and automated result validation. Strong grounding in transformer architectures and FastAPI/AWS deployment.

CORE COMPETENCIES

Programming & ML Frameworks: Python, PyTorch, OpenCV, Hugging Face Transformers, scikit-learn, FastAPI

Generative AI & LLMs: Anthropic Claude API, Tool-Calling / Agentic Workflows, Retrieval-Based Table/Context Selection, Prompt Engineering, LLM Fine-Tuning

Computer Vision & Model Architectures: YOLOv8/v11, YOLO26 (s/m, P2 head), TrOCR, FaceNet, EfficientNet-B0, PaddleOCR, R3D-18

Deployment & DevOps: Docker, AWS EC2, Streamlit, REST API Design

Database: MySQL, PostgreSQL

Networking & System: ngrok, Camera Configuration

Languages & Tools: SQL, Linux, Git

EDUCATION

BCA – BACHELOR OF COMPUTER APPLICATIONS

[JKKN College of Arts and Science](#)

Kumarapalayam, Tamil Nadu

2019 – 2022

PROFESSIONAL EXPERIENCE

TRACK TECH SOLUTIONS

[AI/ML Engineer](#)

Bengaluru, India (Remote)

Apr 2026 – Sep 2026

- Built a **stitch-counting CV system** (FastAPI) using YOLO26s/YOLO26m with P2 heads — separate models for single-needle, double-needle, and overlook stitch types with dedicated scale-point detection; trained on a self-built dataset of **~100K manually point-annotated stitches**.
- Implemented **SAHI sliced inference** for small/edge-stitch recall, class-agnostic duplicate removal (IoU + center-distance), and a scale-anchored band/border filter, plus a row-finder for double-needle/overlook counting and REST endpoints for single, annotated, and **batch (up to 20 images)** inference.
- Built a **natural-language factory reporting chatbot** (FastAPI + SQLAlchemy) translating plain-English questions into safe, read-only MySQL queries via the Anthropic Claude API across a garment production database of **220+ tables** with almost no declared foreign keys.
- Designed a **business-glossary/domain-group table-selection layer** for reliable join-set selection, plus a **SQL safety guard** (SELECT-only, forbidden-keyword filtering, auto-LIMIT) and an empty-result investigator with fallback retries.
- Built a **voice-to-database assistant** using Claude API tool-calling with prompt-based filters to route transcribed voice queries — a hands-free extension of the reporting workflow.

HEXBEE SOFTWARE SOLUTIONS PVT LTD

[AI/ML Engineer](#)

Salem, India (Remote)

Sep 2024 – Apr 2026

- Engineered **2 production CV systems** (ANPR, Face Recognition + Anti-Spoofing) deployed live; curated **100K+ raw images** with custom deduplication and annotation pipelines.
- Fine-tuned Microsoft TrOCR (**93% accuracy**) and built the ANPR pipeline (YOLOv8 + PaddleOCR/TrOCR, Indian-plate regex); two-round fine-tuning (**14K detections + 10K hard negatives**) enabled **~20 FPS** real-time inference on LAN cameras via FastAPI.
- Built a **3-stage face recognition pipeline** (YOLOv11-face → EfficientNet-B0 liveness → FaceNet verification) trained on **10K self-collected anti-spoof images**; deployed on **AWS EC2**.

KEY PROJECTS

- AI Camera Security — Violence Detection:** Fine-tuned R3D-18 (Kinetics-400) on RWF-2000 CCTV data, achieving **81% accuracy / 0.81 F1**; batched GPU inference at **~20 FPS** with a 3-tier Streamlit alert system.
- AI-Powered Image Annotation Tool:** Local-first CVAT-inspired tool (FastAPI + JS) with YOLO pre-annotation, human-in-the-loop review, **multi-format export (YOLO/COCO/VOC/JSONL)**, and in-tool few-shot fine-tuning.
- Self-Persona AI (LoRA Fine-Tuned Career Chatbot):** Fine-tuned Llama-3.2-3B-Instruct via LoRA on **541 hand-built examples** grounded in own project history, with assistant-turn loss masking and early stopping to prevent hallucination; dual-GPU torchrun training on Kaggle T4s, served via a **4-bit quantized** Streamlit app.

CERTIFICATIONS

Machine Learning with Python (IBM) • Data Visualization (IBM) • Data Analysis (IBM) • Deep Learning & Neural Networks with PyTorch (Udemy) • Python Data Structures & Algorithms (Udemy) • Complete Data Science, ML, DL & NLP Bootcamp (Udemy) • SQL Basic & Python Basic (HackerRank)